

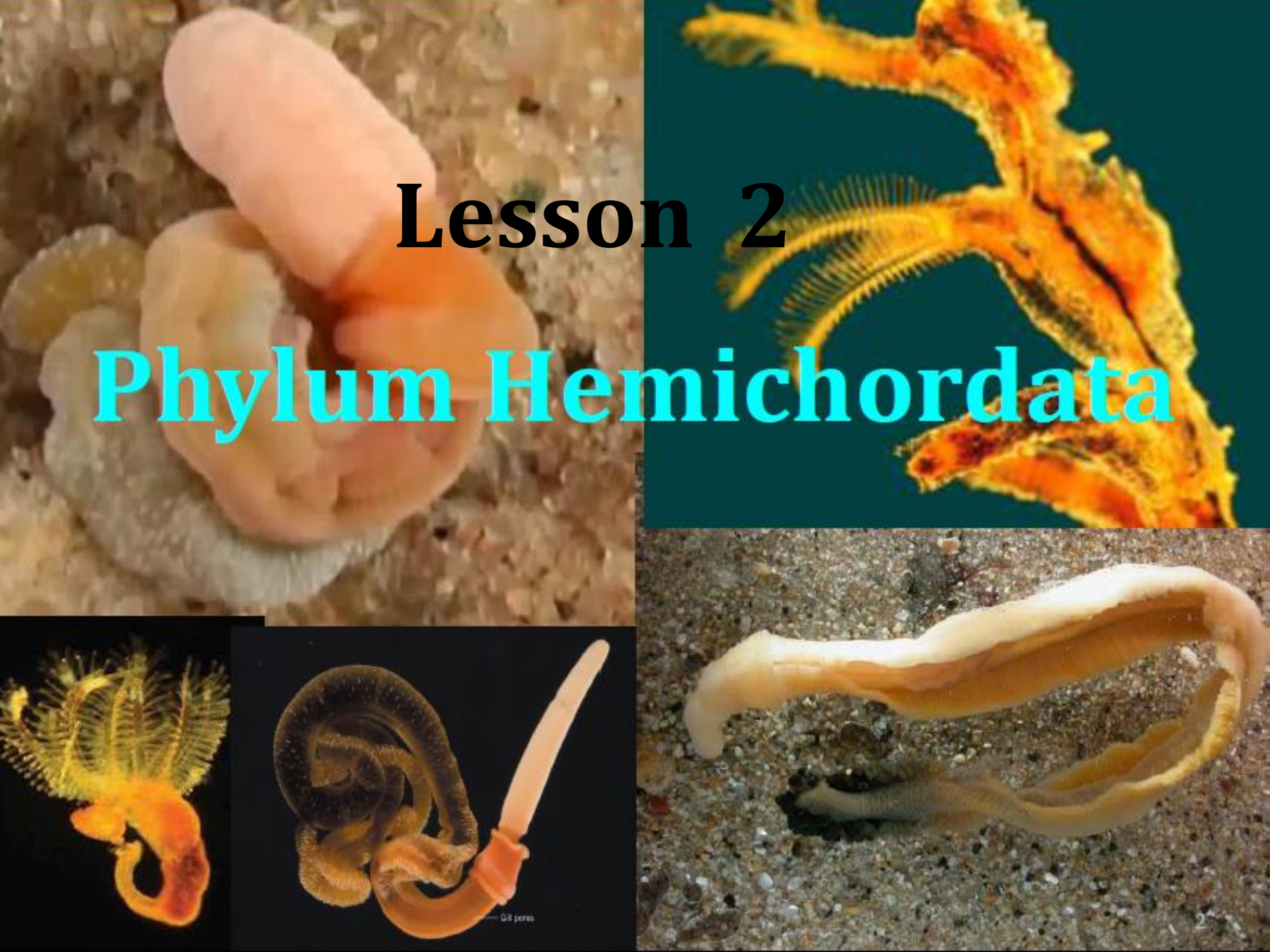


ZOO 1302 Vertebrate Diversity

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Lesson 2

Phylum Hemichordata



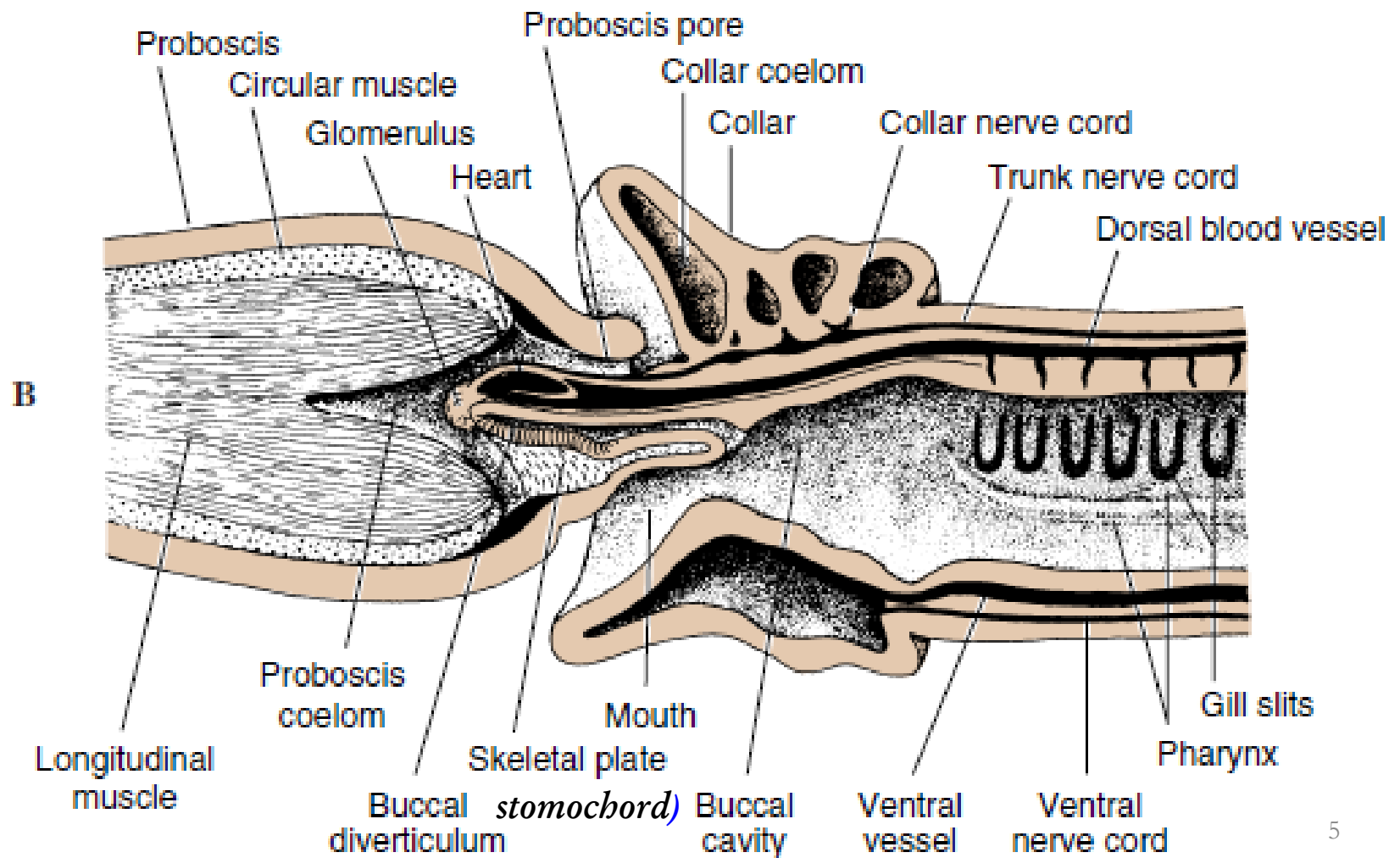
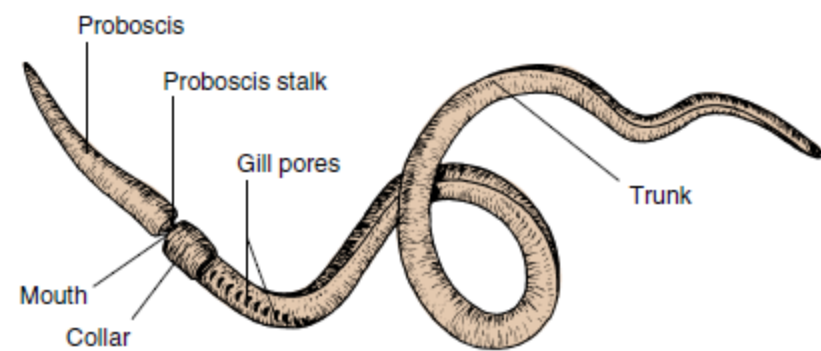
Learning Outcomes

At the end of the lesson you would be able to describe the

- characteristics of phylum Hemichordata.
- general characteristics of *Balanoglossus*.
- phylogeny of Hemichordates.

Phylum Hemichordata

- Hemichordata - *Hemi* – half + *chordata* – string, cord.
- These marine animals formerly considered a subphylum of chordates based on their possession of gill slits and a rudimentary notochord.
- Hemichordate notochord – is a buccal diverticulum and not homologous with the chordate notochord – *stomochord* (mouth-cord).
- Hence, hemichordates are given the rank of a separate phylum.



Characteristics of Phylum Hemichordata

- Body divided into **proboscis**, **collar**, and **trunk**
- Buccal diverticulum in posterior part of proboscis
- Bilaterally symmetrical.
- Soft bodied; wormlike or short and compact with stalk for attachment
- Triploblastic

■ Two classes

1. Class-Enteropneusta (acorn worms)

- Eg. *Balanoglossus*
- free-moving and of burrowing habits

2. Class: Pterobranchia (pterobranchs)

- Eg. *Rhabdopleura*
- sessile, mostly colonial, living in secreted tubes

- Single coelomic pouch in proboscis, but paired pouches in collar and trunk
- Ciliated epidermis
- Digestive system complete
- Longitudinal and circular muscles in body wall in some
- ventral nerve cords, with a ring connective in the collar;
- Some species with hollow dorsal nerve cord
- Sensory neurons in proboscis likely function in chemoreception

- A single **glomerulus** connected to blood vessels may have excretory function - metanephridium
- Respiratory system - **gill slits** (few or none in pterobranchs)
- Gill slits connecting the pharynx with outside
- **Circulatory system** - dorsal and ventral vessels and dorsal heart

- *Asexual reproduction*

- Pterobranchs - Colonies form by asexual budding
- Enteropneusts - reproduction by fragmentation

- *Sexual reproduction*

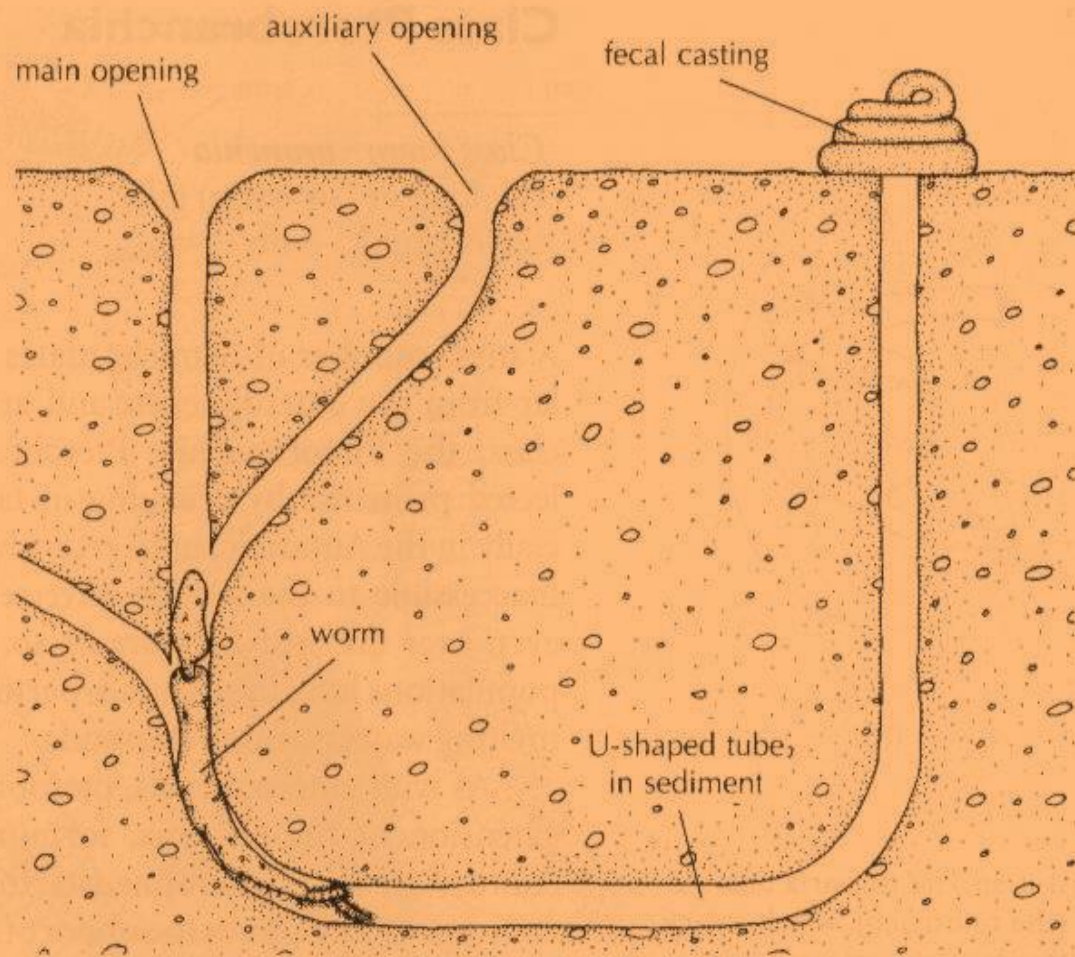
- Enteropneusta - sexes separate in with gonads projecting into body cavity
- tornaria larva in some Enteropneusta

Class-Enteropneusta (acorn worms)

Eg. *Balanoglossus*

- *Balanoglossus* is world-wide or cosmopolitan in distribution.
- Sluggish worm like animals
- Habitat - Exclusively marine.
- Live in burrows in mud or sand flats of intertidal zone – shallow waters
- Tubicolous - living in U-shaped burrows excavated in the sandy bottom.
- It burrow slowly by soft proboscis.

- The walls of the tube are lined with mucus secreted by the mucous gland of the animal.
- The burrows are open at both ends, and spiral coils of faeces like the castings of earthworms may be seen at the posterior opening.
- In its burrow, *Balanoglossus* lies in a twisted condition but its anterior and posterior extremities are straight.
- The animal moves in its burrow with the help of cilia present all over the body.

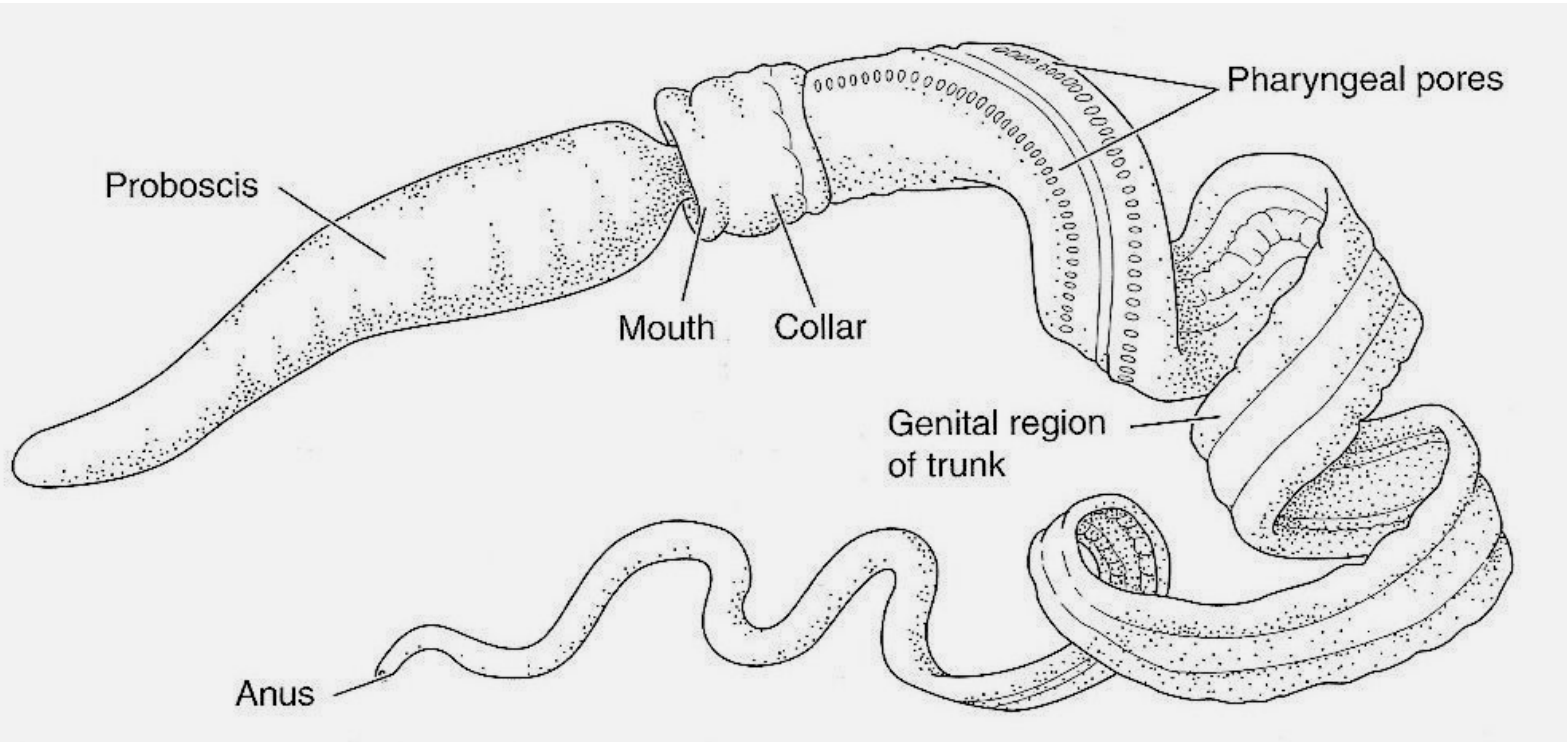


External Morphology of *Balanoglossus*

- The body is soft, elongated, cylindrical, richly ciliated all over and covered with mucus.
- The length of animal varies from 2 cm to 2.5 meters.
- Most forms are drab coloured, though reddish tints are present and several species are luminescent due to mucus.
- They have an offensive odour – defense mechanism
- The body is bilaterally symmetrical.

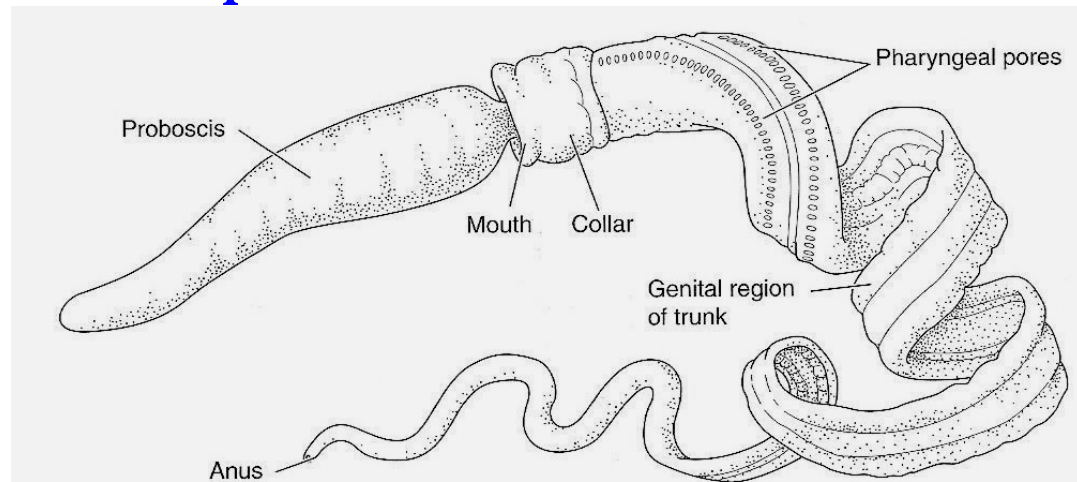


- Body is divided into three regions.
 - a) proboscis – protosome
 - b) collar – mesosome
 - c) trunk – metasome
- It is also called – tongue worm due to the resemblance of the proboscis and genital wing to ox tongue



a) Proboscis

- The proboscis forms the anterior part of the body and is either rounded or conical in shape.
- It is continued posteriorly into a short, narrow neck or proboscis stalk.
- The proboscis is hollow and has thick muscular walls.
- Its cavity opens to the outside by means of a small opening called the proboscis-pore.

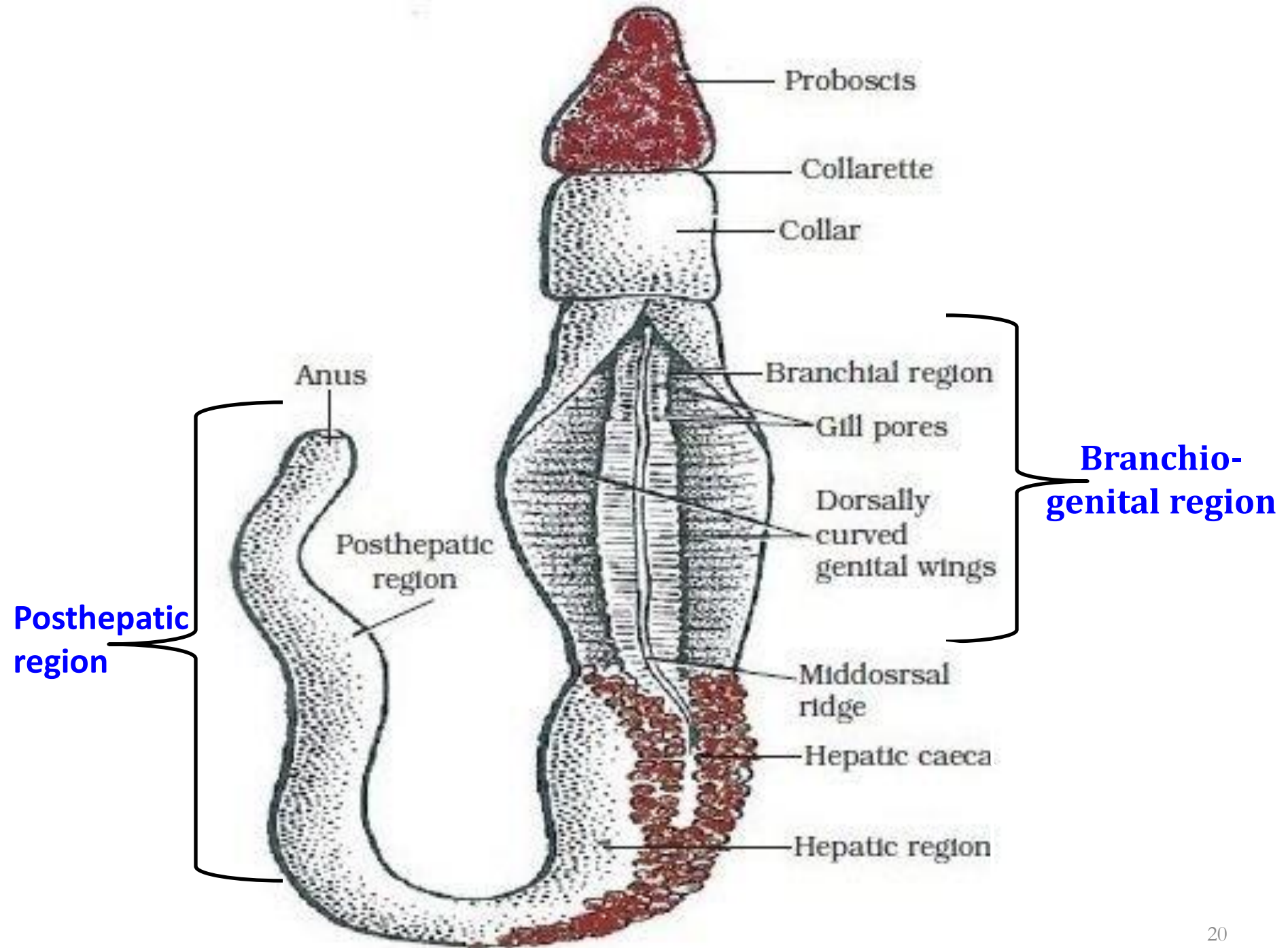


b) Collar

- The collar lies posterior to the proboscis and anterior to the trunk.
- It is a short cylinder usually about as wide as long and mostly shorter than the proboscis.
- The funnel-like anterior part of the collar, the collarete, embraces the proboscis stalk and usually also the posterior part of the proboscis.
- Posteriorly the collar is sharply demarcated from the trunk by a circular indentation (notch).
- The collar is muscular and possesses two coelomic cavities.

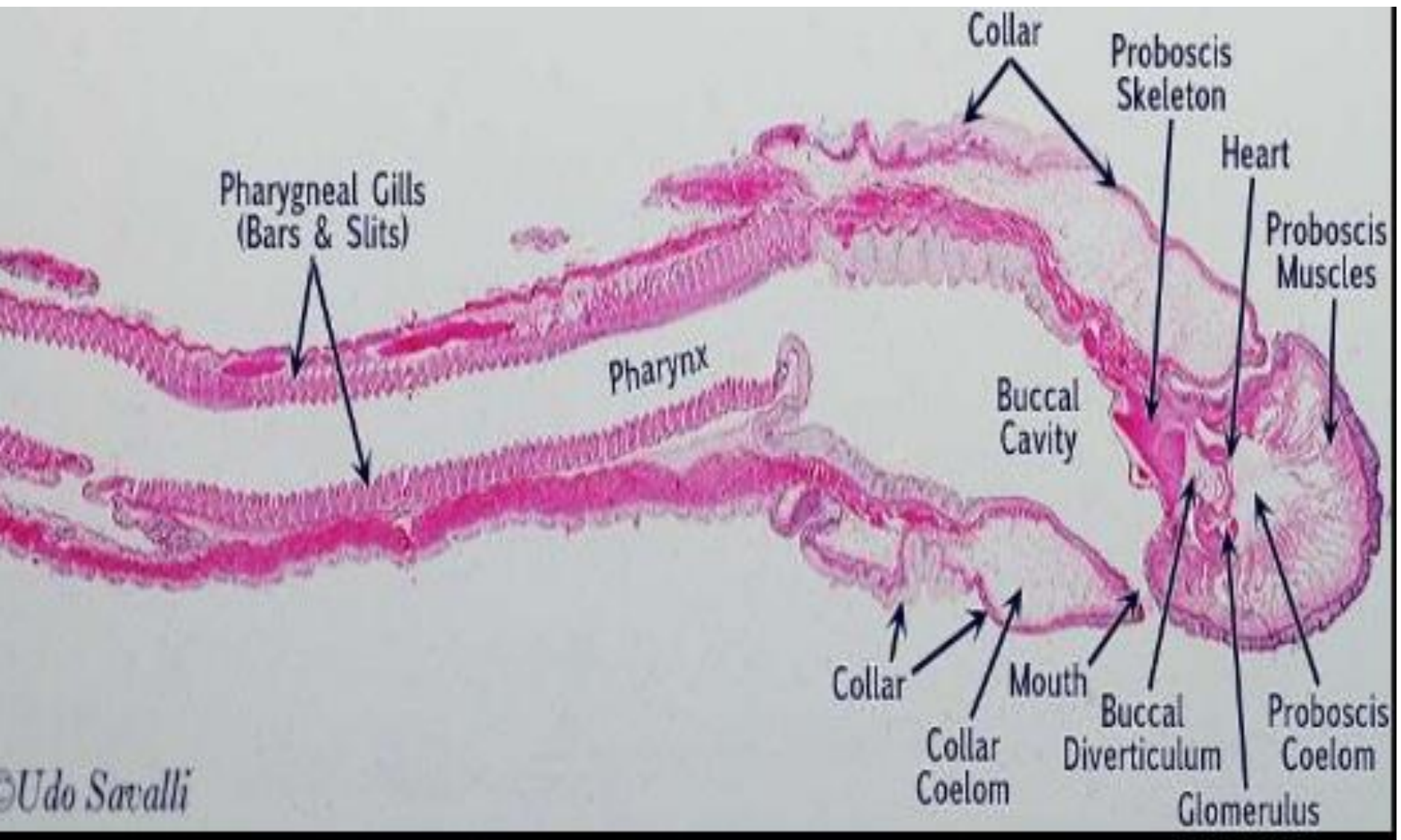
c) Trunk

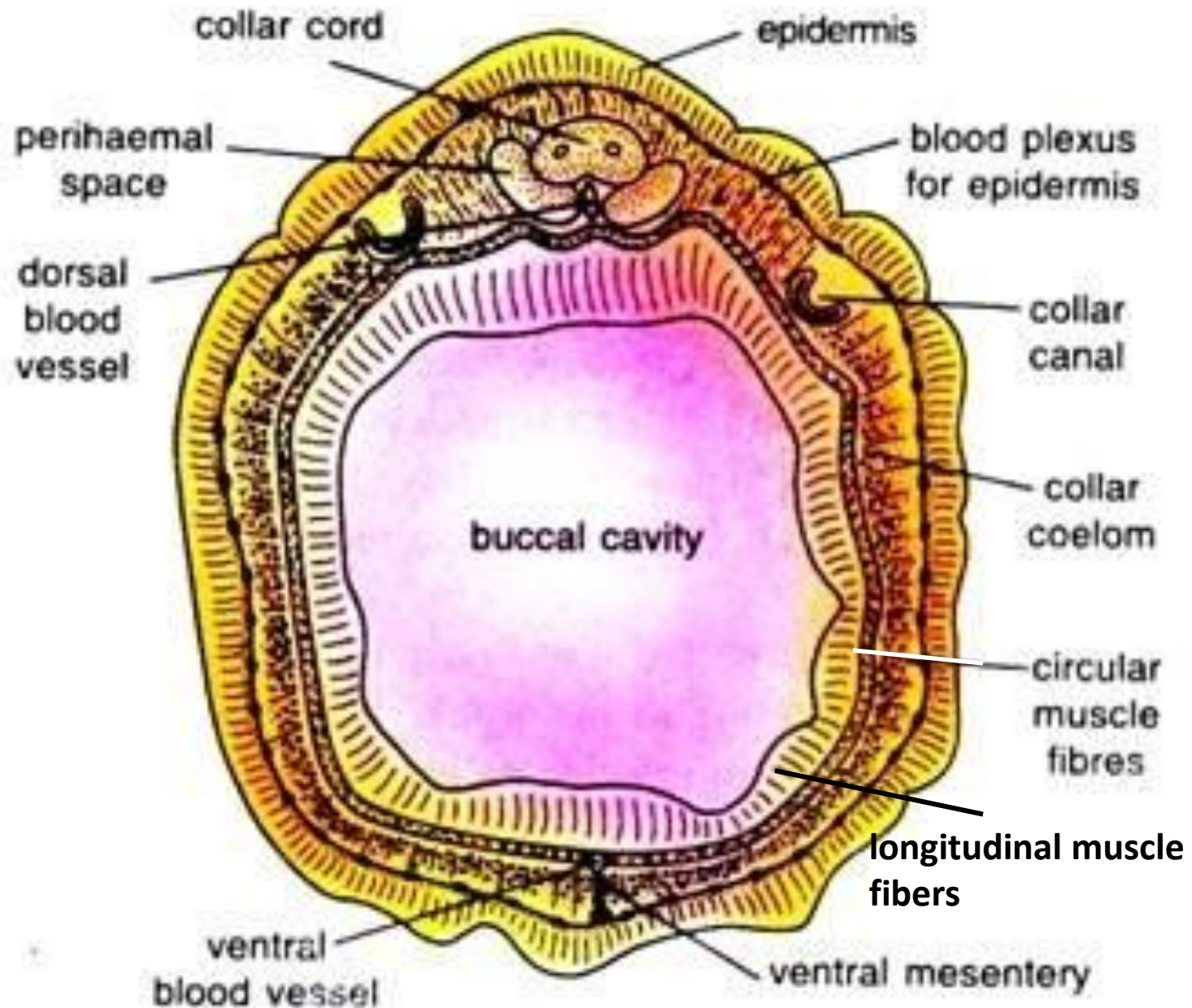
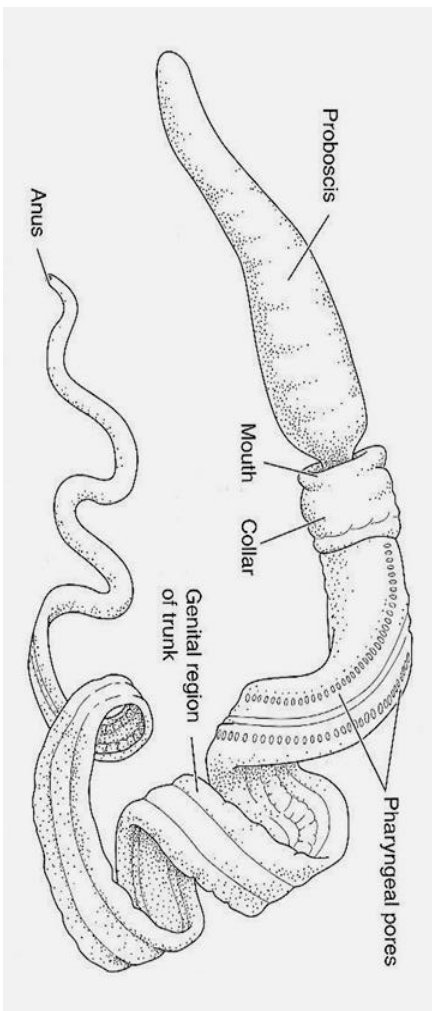
- The trunk is the elongated posterior part of the body.
- It is somewhat flat and annulated on the surface.
- It has a mid-dorsal and a mid-ventral longitudinal ridge.
- The trunk is divisible into three parts,
 1. branchio-genital region - anterior
 2. hepatic region - middle
 3. abdominal or post-hepatic region- posterior



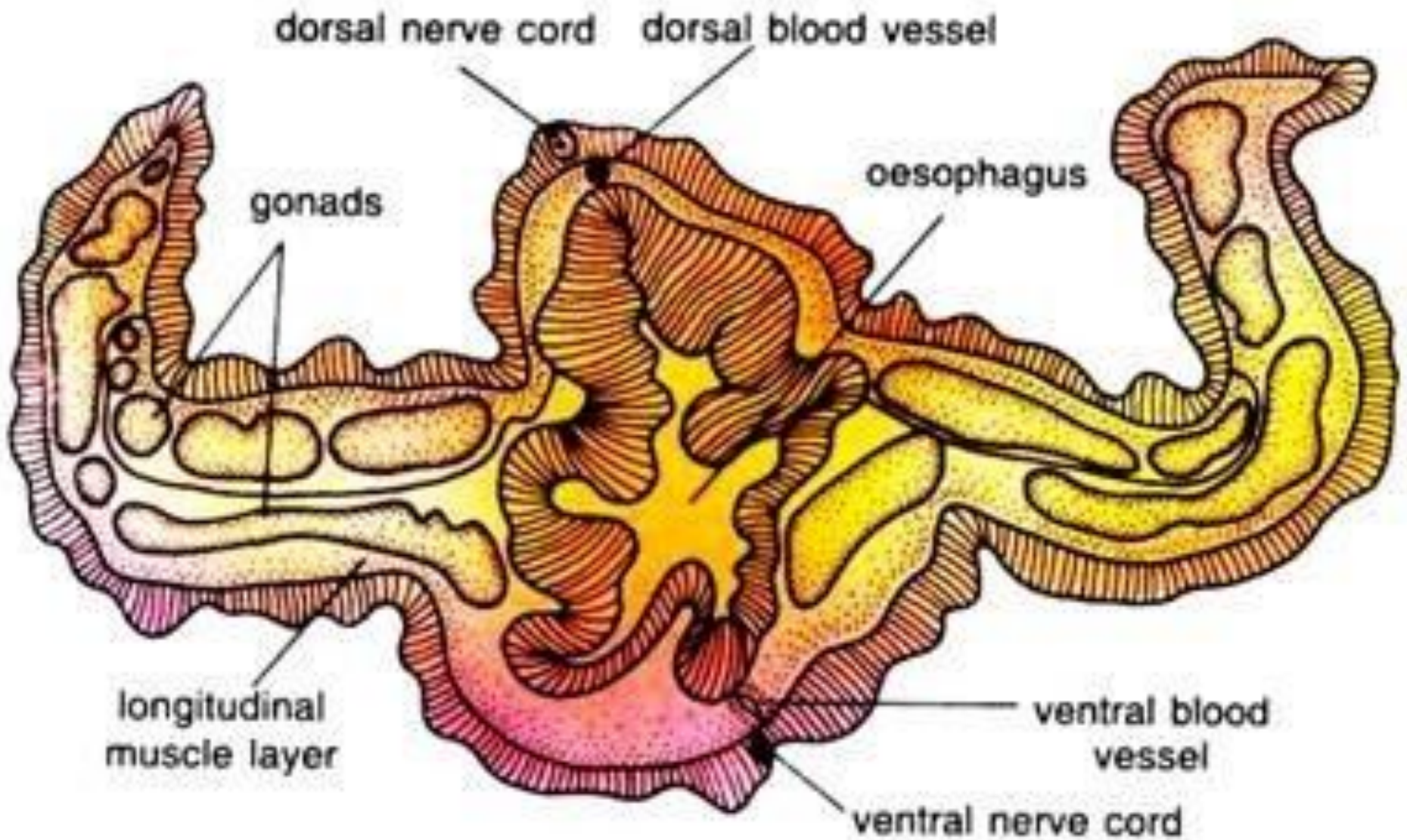
1. Branchio-genital region

- On the dorsal of the branchio-genital region of the trunk is a double row of small pores the branchial apertures.
- Each row is situated in a long furrow.
- A pair of longitudinal genital ridges or genital wings extends throughout a considerable part of the body behind and in the region of the branchial apertures.
- In these genital ridges, gonads are situated.
- In some genera, the genital ridges are so prominent that they form a pair of wing-like lateral folds, the genital wings, but in other genera folds are absent.





***Balanglossus* T.S.of collar**



***Balanglossus* C.S. through genital wing**

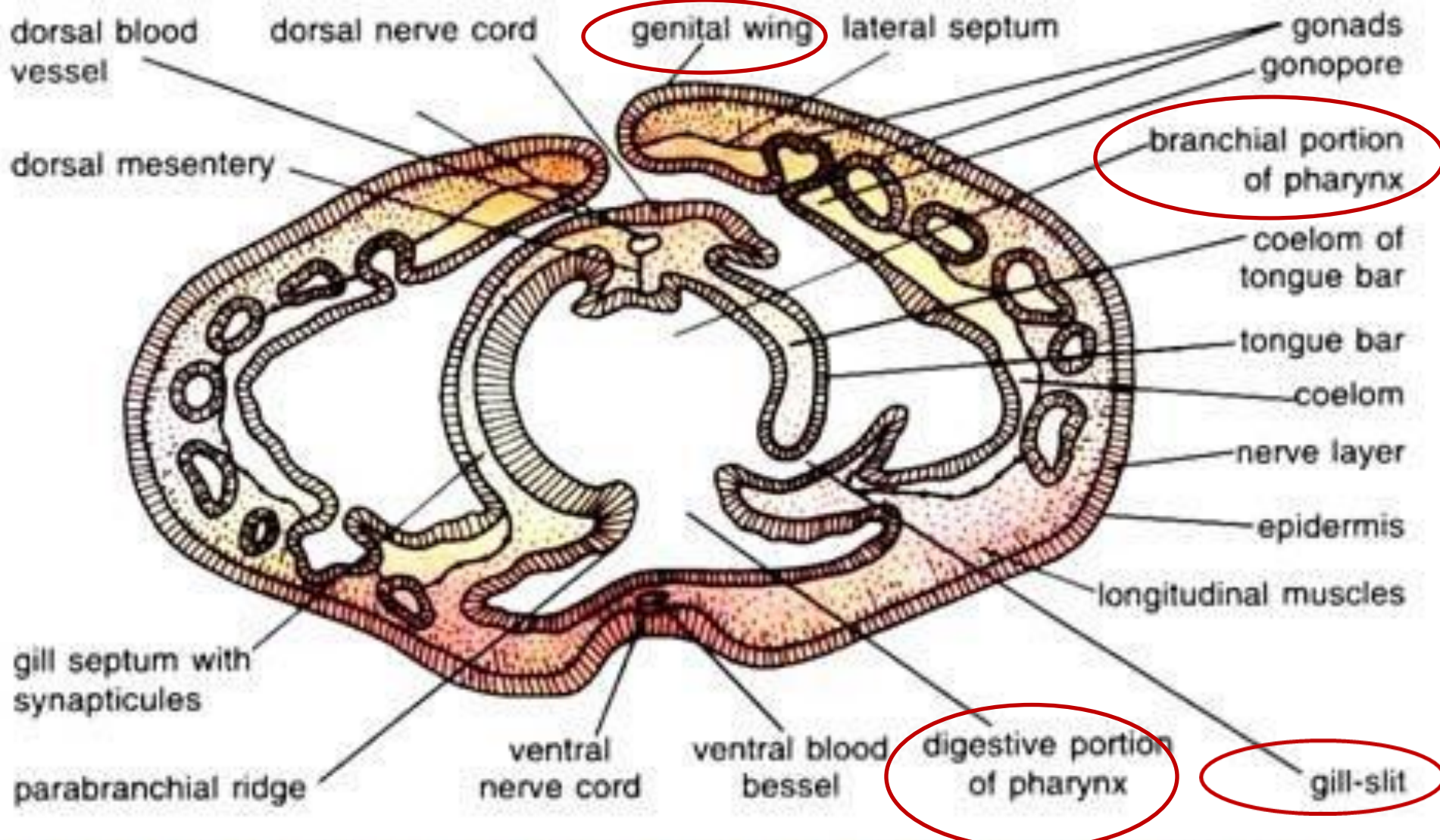


Fig. 88.9. *Balanoglossus*. T.S. through pharyngeal or branchiogenital region.

***Balanglossus* T.S. through branchiogenital region**

- A pair of longitudinal genital ridges or genital wings extends throughout a considerable part of the body behind and in the region of the branchial apertures.
- In these genital ridges, gonads are situated.

2. Hepatic region

- Hepatic region is marked externally with irregular elevations due to sacculations produced by projecting hepatic caeca of the intestine.

3. Post hepatic region

- The post hepatic or abdominal region is longest and cylindrical.
- It tapers gradually and has a terminal anus.

Endoskeleton

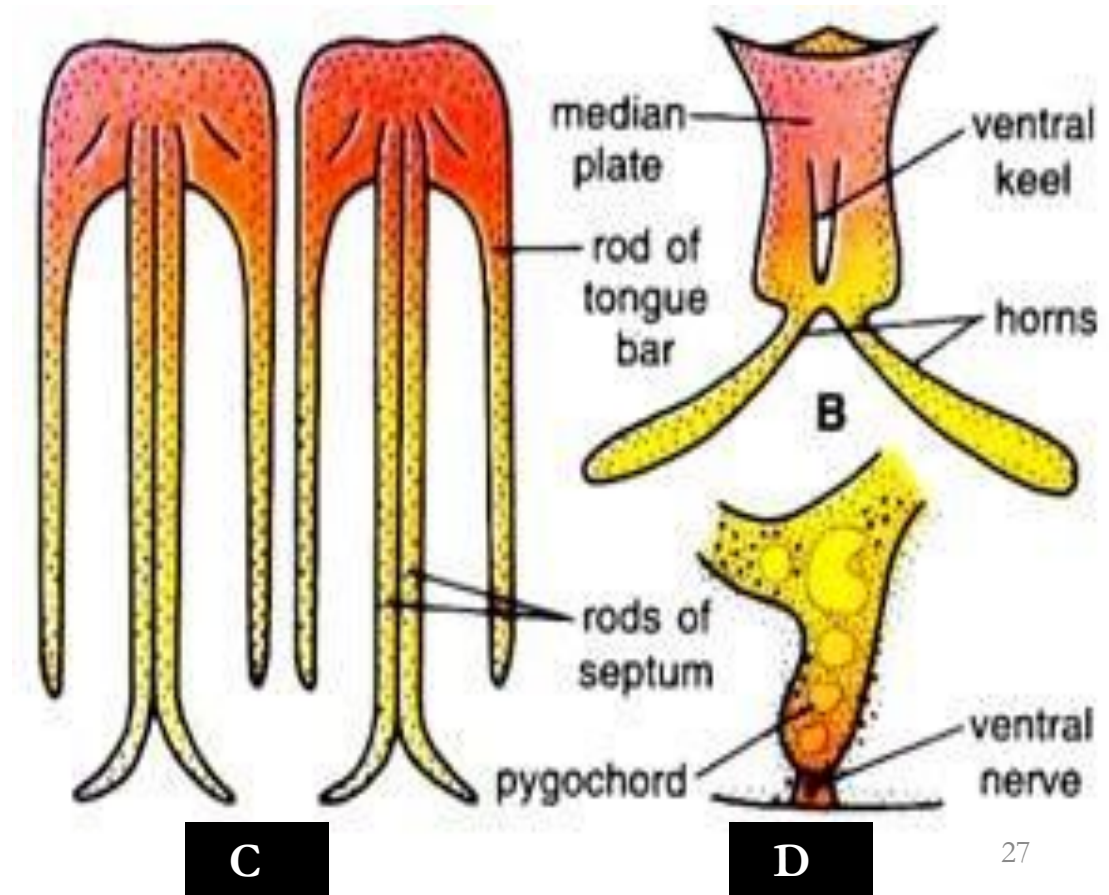
- In Balanoglossus there is no definite endoskeleton.
- But there are four structures of a supporting nature.

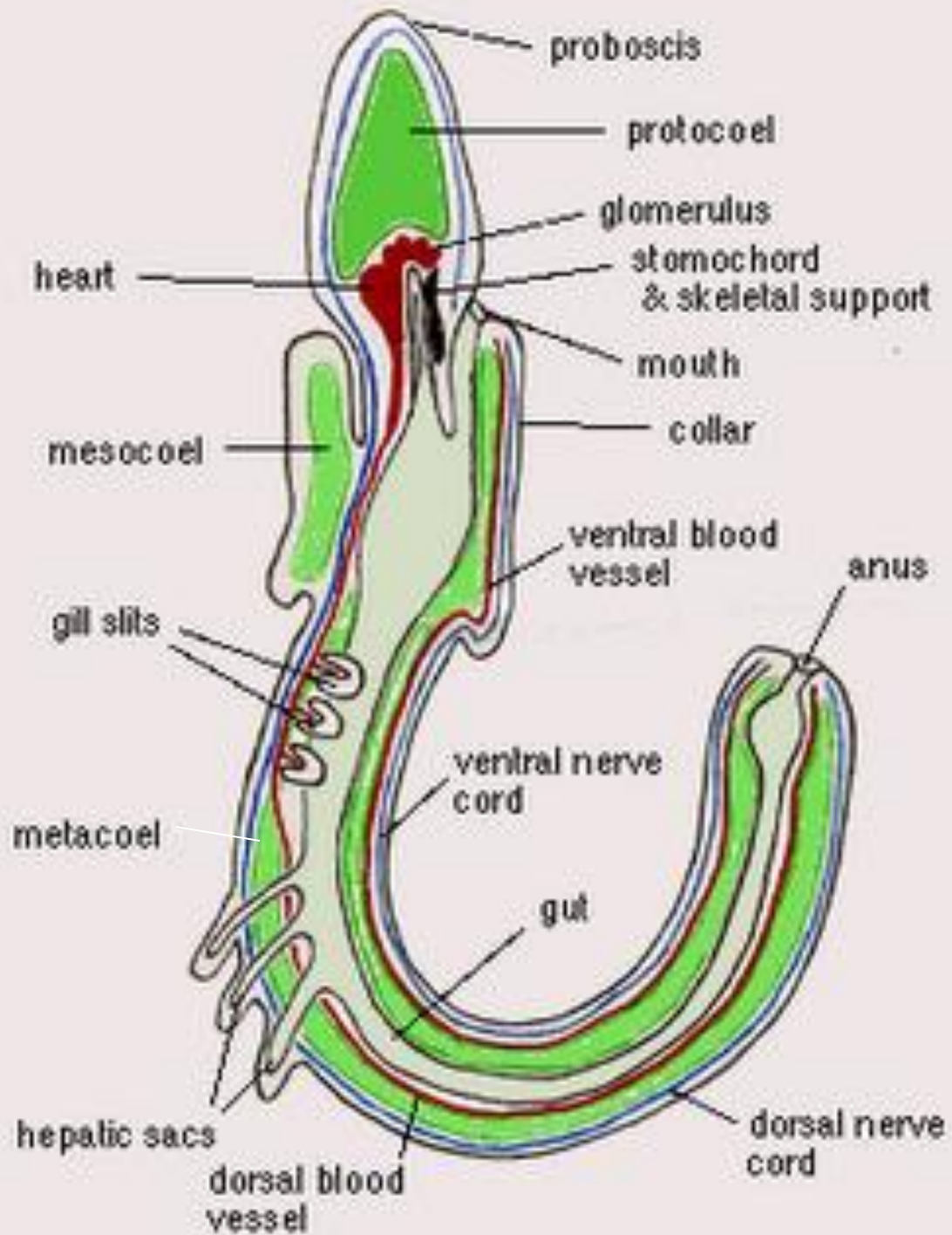
A) Buccal diverticulum (stomochord),

B) Proboscis skeleton,

C) Branchial skeleton,
and

D) Pygochord.

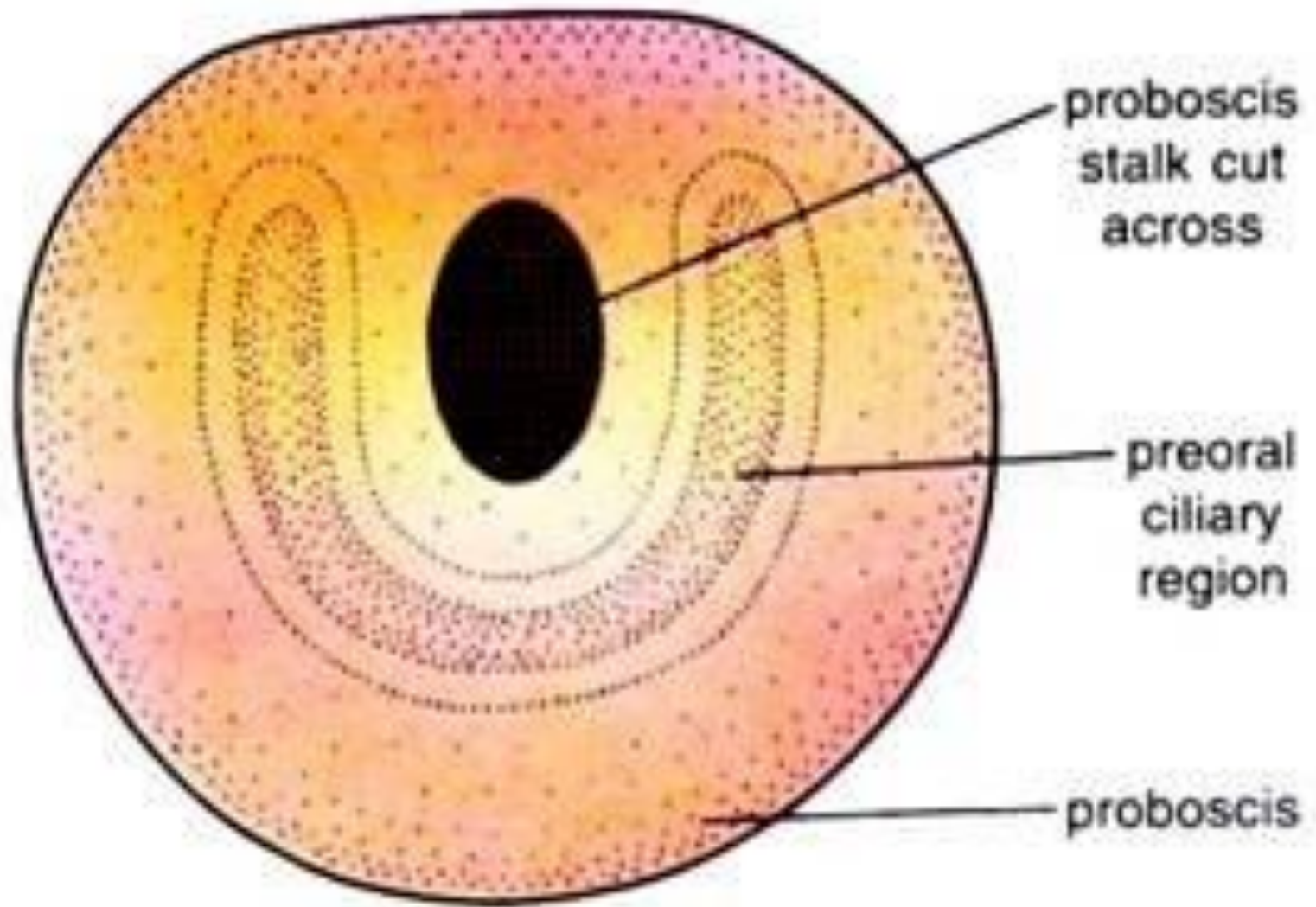




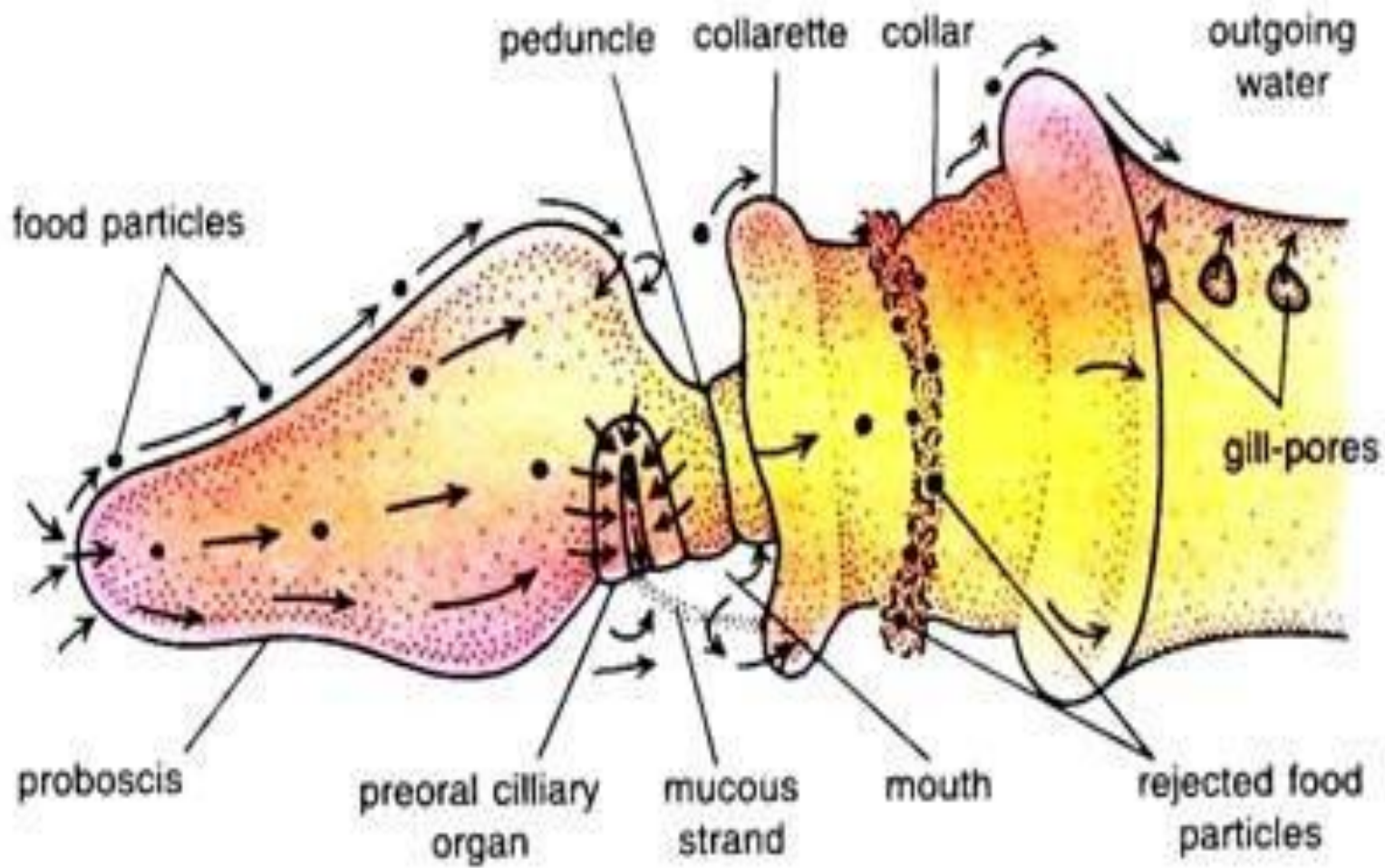
Food and Feeding of Balanoglossus

- Balanoglossus is a “ciliary feeder”.
- Its food comprises microscopic organisms and minute organic particles present in water and mud or bottom sand in which it makes its burrows.
- The lateral cilia lining the gill-slits set up a current of water directed backward.
- This water current enters through the mouth, takes its course through the buccal tube, pharynx, gill-slits and branchial sacs, and leaves through gill-pores.

- This is called respiratory-cum-food current.
- Some food particles directly enter the mouth with this current, while some come in contact with the proboscis and are entangled in the mucus that covers it.
- The mucus is secreted by the gland cells of the proboscis epithelium.
- Cilia covering the proboscis direct the mucous string, containing food particles towards the preoral ciliary organ at the base of the proboscis.
- From here the mucous string is passed back into the mouth by the action of the proboscis cilia, assisted by the main water current entering the mouth.

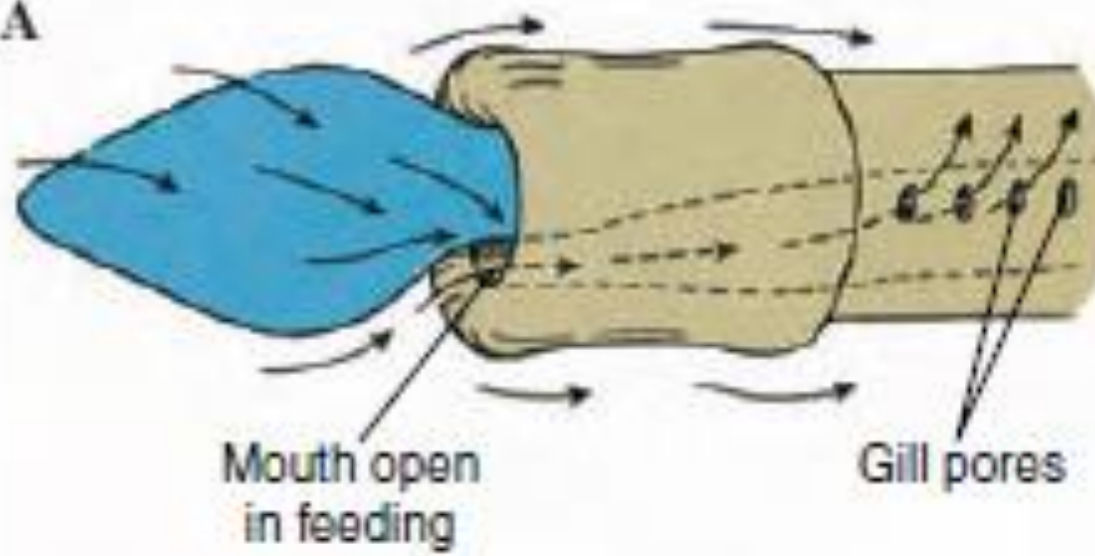


***Balanoglossus* – Posterior view of proboscis showing the preoral ciliary organ**

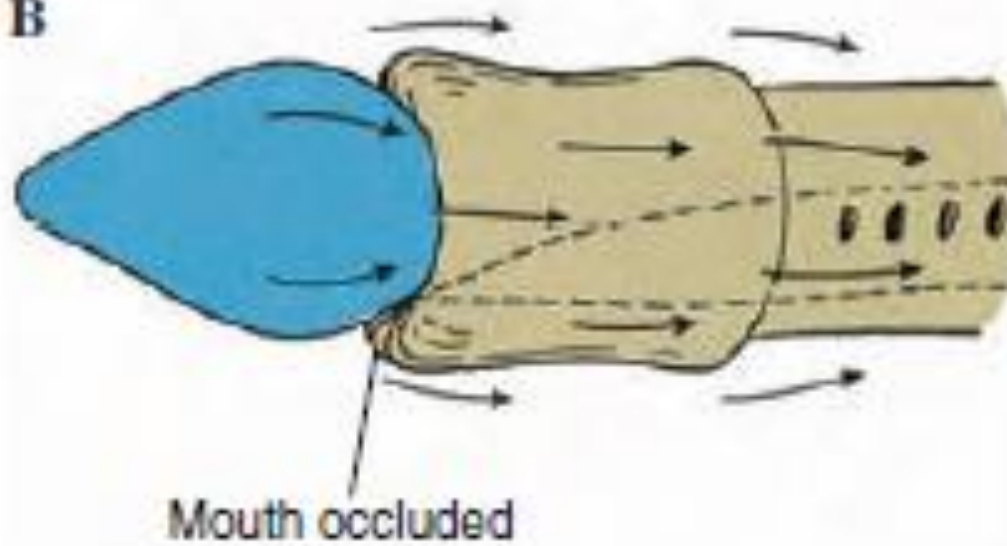


***Balanoglossus* - feeding currents**

A



B



Excretory system

- Excretory organ is glomerulus or proboscis gland
- Lies in front of the central sinus and projecting into the proboscis coelom

Nervous system

- Very primitive type - resembles coelenterates and echinoderms
- Consists primarily of an epidermal plexus or a layer of nerve cells and nerve fibres lies just below the epidermis
- Threadlike processes of the epidermal cells contribute to the network or nerve net

Sense organs

- Poorly developed
- Epidermis are numerous neurosensory cells which are connected to the nerve net – numerous on proboscis

Respiratory System

- Respiratory organs of *Balanoglossus* comprise
 - a) Branchial portion of pharynx bearing gill-slits
 - b) Branchial sacs that open out through gill-pores

Blood vascular system

- Consists of closed vessels, lacunar spaces and a definite pulsating organ , the heart
- Blood is colorless and has no or very few corpuscles
- Has no respiratory pigment
- Two main longitudinal vessels, the dorsal and ventral vessels – running along the length of the body.

Reproduction

- Sexes are separate.
- Indistinguishable externally except in case of the colour of the ripe gonads shown through the body wall in the living animal.
- Shows asexual and sexual reproduction.

Asexual reproduction

- ✓ It has great power of regeneration, small pieces are constricted from the posterior end, each of which regenerates into a complete individual.

Sexual reproduction

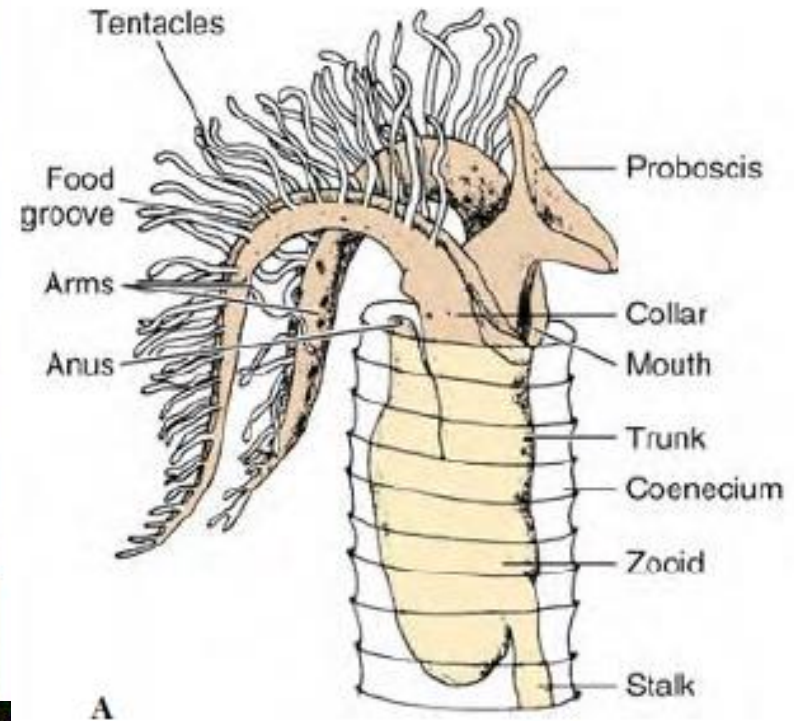
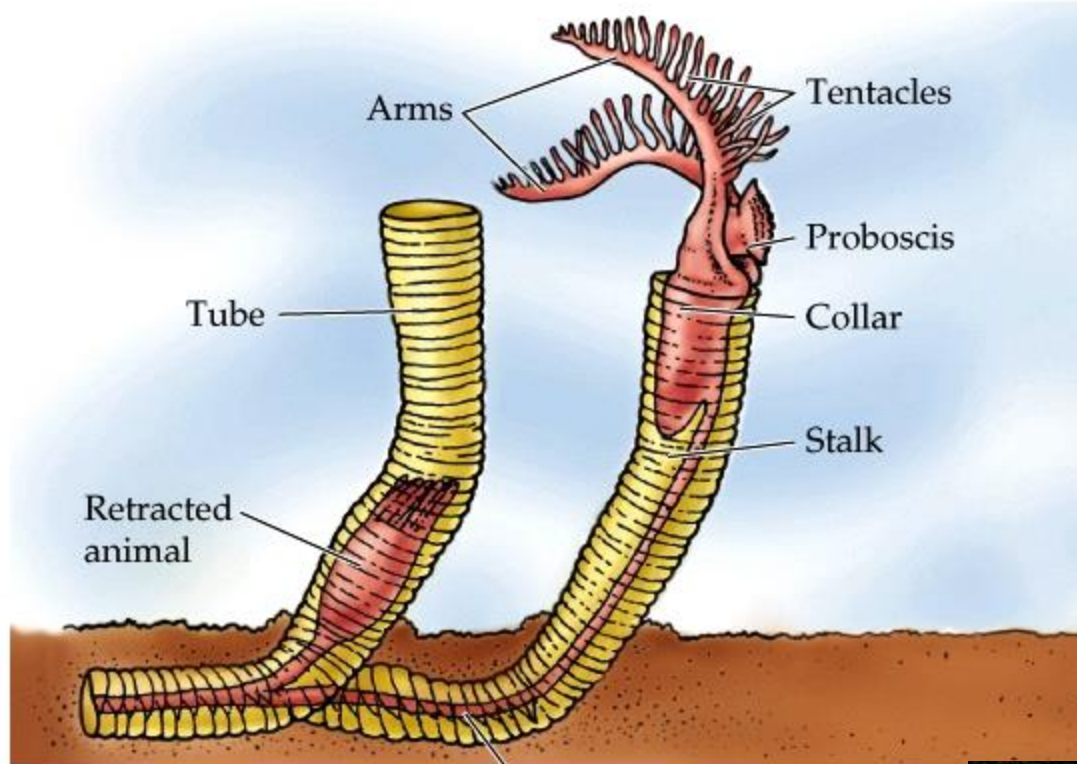
- ✓ External fertilization takes place in water.
- ✓ Embryo develop to a transparent oval shaped tornaria larva.

Class: Pterobranchia (pterobranchs)

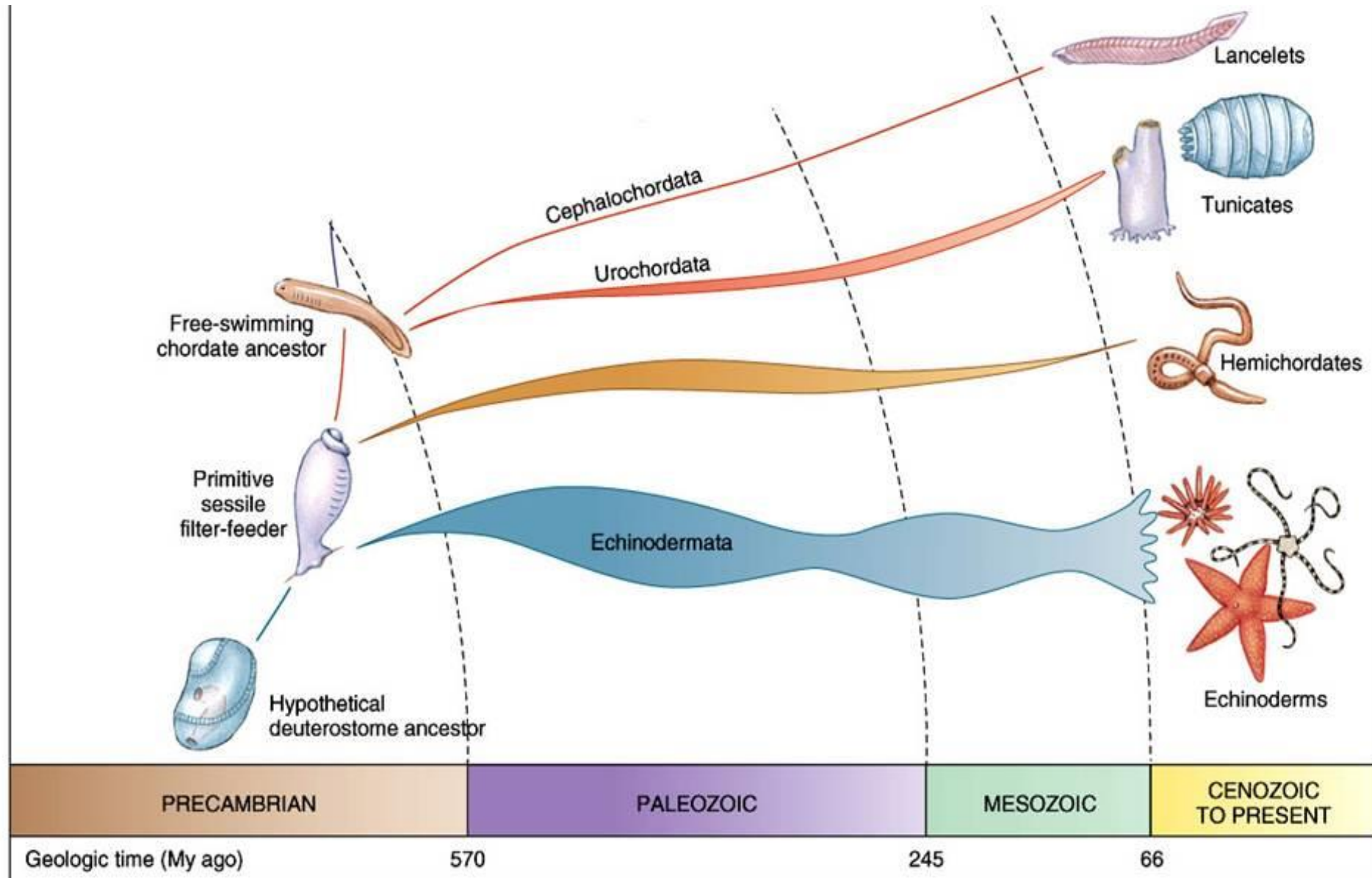
- Pterobranchs are small animals, usually within a range of 1 to 7 mm in length.

Eg. Rhabdopleura

- Individuals remain together to form a colony of zooids
- Zooids are connected by tissues called stolon.
- Zooids are enclosed in secreted tubes.
- Collar bears two arms.
- No gill slits present.



Phylogenetic Relationship between Echinoderms, Hemichordates & Chordates

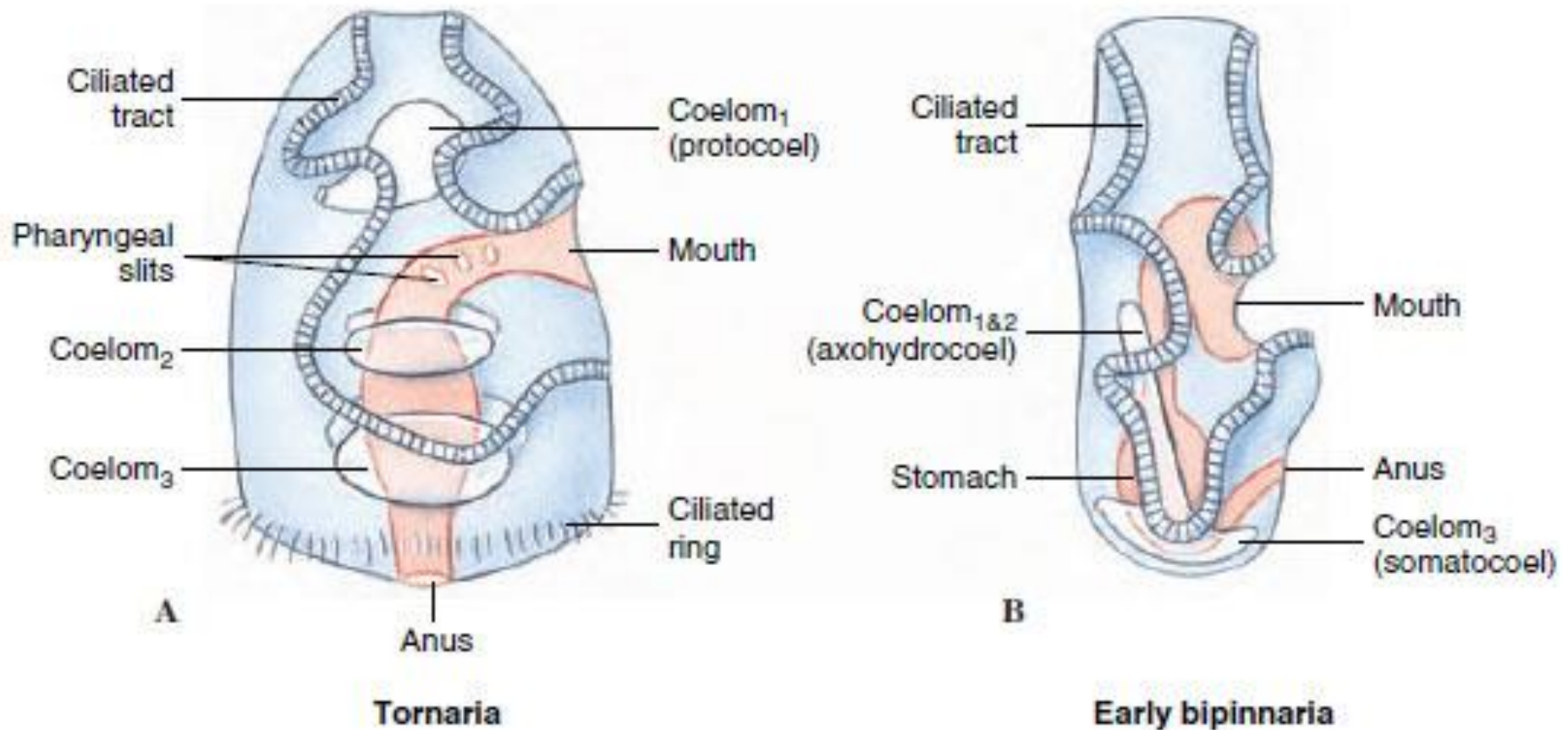


Phylogeny of Hemichordates

- Hemichordates share characteristics with both echinoderms and the chordates.

With chordates they share

- ***Gill slits*** - serve primarily for filter feeding and secondarily for breathing as they do in some protochordates.
- A short dorsal, some what ***hollow nerve cord*** in the collar region may be homologous to the nerve cord of the chordates.
- Their early embryogenesis is remarkably like that of echinoderms and the early larvae is almost identical with the bipinnaria larva of asteroids.



Comparison of a hemichordate tornaria larva (A) to an echinoderm bipinnaria larva (B)

Differences between Hemichordates and Chordates

- Notochord -
- Nervous system -
- Circulatory system :
- Blood pigment -
- Transportation of respiratory gases and metabolic wastes -
- Postanal tail -



The image shows two symmetrical, translucent, feathery structures, likely biological in origin, possibly larval appendages or specialized cells. They are positioned on either side of the central text. The structures have a dense, fan-like top and a more complex, branching base. The background is solid black.

The end